



Time: 5 minutes

Outcome: Calculate output per worker and interpret productivity changes.

Difficulty: Intermediate

Productivity Measurement

THE CORE IDEA

Labor productivity measures output per worker or per hour worked. For example, 2,400 units produced in 300 labor hours means $2,400 / 300 = 8$ units per hour. Productivity compares output with the labor input used to produce it, so changes in both output and labor matter. A country can consume more when its workers produce more output per worker.

HOW TO RECOGNIZE IT

- Measure labor productivity as real output per worker or per hour worked.
- Use the same output and labor units when comparing periods.
- An increase in total output need not raise productivity if labor input also rises.

WATCH OUT

Do not confuse total output with output per unit of labor. Productivity compares output with the labor input used to produce it, so changes in both output and labor matter.

WORKED EXAMPLE: MEASURING LABOR PRODUCTIVITY

A workshop produces 2,400 units using 300 labor hours, so productivity is $2,400 / 300 = 8$ units per hour. Next month it produces 2,700 units using the same 300 hours: 9 units per hour. Productivity rises by $(9 - 8) / 8 \times 100 = 12.5\%$. More output per labor hour, not simply more total hours, is the improvement.

CHECK YOURSELF

A business uses more labor hours but produces the same amount of output. What happens to labor productivity?